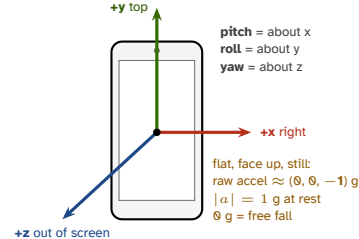


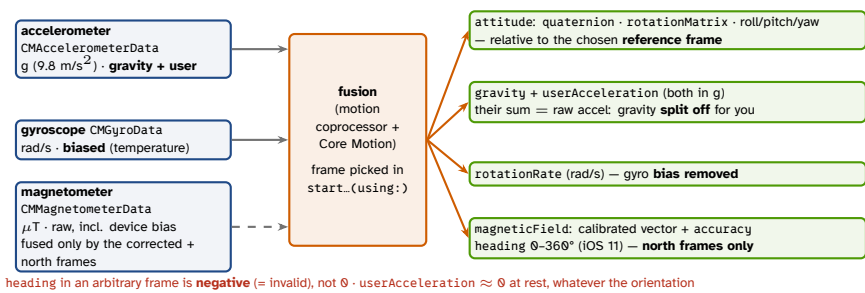
**In one line:** Core Motion turns the **accelerometer** (g, gravity included), **gyroscope** (rad/s, biased) and **magnetometer** ( $\mu\text{T}$ , uncalibrated) into **device motion** by **sensor fusion**: attitude, gravity split from user acceleration, bias-free rotation, calibrated field. One CMMotionManager per app; the motion coprocessor also logs steps and activity **while your app is not running** (7 days).

#### Device axes (portrait)



rotation: right-hand rule, rad/s  
Euler angles: radians,  $-\pi \dots \pi$

#### Raw sensors → sensor fusion → CMDeviceMotion



#### CMMotionManager

- **One instance per app** (Apple: several “can affect the rate at which the system receives data”). Keep a strong reference — a deallocated manager delivers nothing.
- Four services, each with `isAvailable / isActive`, an `updateInterval` in **seconds** ( $1.0/60$ ), and `start/stop`: accelerometer, gyro, magnetometer, **deviceMotion**. Max rate is hardware-dependent, “usually at least **100 Hz**”; asking for more silently gives the max.
- **Push** = `start...(to: queue) { }`: every sample to an `OperationQueue`. **Pull** = `start...()` with no handler, then read `mm.deviceMotion` once per frame (`CADisplayLink`) — Apple’s pick for games.
- Battery: sensors + fusion run only while started — lowest rate that works, stop off screen. Updates stop when the app is suspended (no “motion” background mode).
- **Prefer deviceMotion** over raw (see picture); `magneticField.accuracy.uncalibrated...high`; `showsDeviceMovementDisplay` lets iOS ask the user to move the device to calibrate. `attitude.multiply(byInverseOf: ref) = rotation` since a reference pose (“recentre”).

#### Example — fused motion, push, one manager

```
final class Motion { // ONE per app, kept alive
  let mm = CMMotionManager()
  private let queue = OperationQueue() // not .main at 100 Hz
  func start(_ onTilt: @escaping (CMAcceleration) -> Void) {
    guard mm.isDeviceMotionAvailable else { return } // Simulator: false
    mm.deviceMotionUpdateInterval = 1.0 / 50 // SECONDS
    let f: CMAttitudeReferenceFrame = .MagneticNorthZVertical
    guard CMMotionManager.availableAttitudeReferenceFrames()
      .contains(f) else { return }
    mm.startDeviceMotionUpdates(using: f, to: queue) { m, error in
      guard let m else { return }
      onTilt(m.gravity) // (0,0,-1) when flat
      let jolt = m.userAcceleration // gravity already removed
      let q = m.attitude.quaternion // x, y, z, w
      let deg = m.heading // valid in north frames
    }
  }
  func stop() { mm.stopDeviceMotionUpdates() }
}
```

#### Authorization

NSMotionUsageDescription is **required** by `CMPedometer`, `CMMotionActivityManager`, `CMSensorRecorder`, `CMHeadphoneMotionManager` — missing → **crash**, not a silent denial. `CMAuthorizationStatus` (iOS 11): `.notDetermined`, `.restricted`, `.denied`, `.authorized`. Raw `CMMotionManager`: no status API until `authorizationStatus()` appeared in the iOS 27.2 **beta** docs (no discussion yet).

#### Attitude reference frames — Z is always vertical

| frame                                     | X axis points to                                      |
|-------------------------------------------|-------------------------------------------------------|
| <code>xArbitraryZVertical</code>          | arbitrary — the default                               |
| <code>xArbitraryCorrectedZVertical</code> | arbitrary; magnetometer fixes long-term yaw, more CPU |
| <code>xMagneticNorthZVertical</code>      | magnetic north — compass                              |
| <code>xTrueNorthZVertical</code>          | geographic north — needs location                     |

#### Higher-level managers — use these before raw data

|                                       |                                                                                                                                                                                                                                                     |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>CMPedometer</code>              | iOS 8. Steps, distance (m), floors up/down, currentPace ( <b>s/m</b> ), currentCadence (steps/s). <code>queryPedometerData(from:to:)</code> = <b>7 days</b> of history; <code>startUpdates(from:)</code> live; <code>isAvailable</code> per metric. |
| <code>CMMotionActivityManager</code>  | iOS 7. stationary walking running automotive cycling unknown + confidence low/medium/high; live or <code>queryActivityStarting</code> ( <b>7 days</b> ).                                                                                            |
| <code>CMAltimeter</code>              | iOS 8. Barometer: relativeAltitude (m since first event), pressure ( <b>kPa</b> ). <b>iOS 15</b> : absolute altitude above sea level + accuracy.                                                                                                    |
| <code>CMHeadphoneMotionManager</code> | iOS 14. Head tracking from spatial-audio AirPods (attitude, rotation, user accel); connect/disconnect delegate.                                                                                                                                     |
| <code>CMBatchedSensorManager</code>   | watchOS 10, Series 8 / Ultra, in a <b>HealthKit workout</b> : accel <b>800 Hz</b> , device motion <b>200 Hz</b> , one batch per second — golf swings.                                                                                               |
| <code>CMFallDetectionManager</code>   | watchOS 7.2, Series 4+: fall events, app woken in background; <b>Apple-granted entitlement</b> .                                                                                                                                                    |

**SensorKit** (iOS 14): research studies only — an entitlement + an approved study; never a product feature.

#### Traps

- ! “Subtract 1 g from z” works only lying flat — use `userAcceleration` (fused with the gyro).
- ! Interval is **seconds**, not Hz:  $= 60$  is one sample a minute.
- ! Handler on `.main` at 100 Hz floods the UI thread — background queue, hop to main.
- ! Euler: **gimbal lock** at pitch  $\pm 90^\circ$  — use the quaternion.
- ! Steps while the app was closed: no background mode — `queryCMPedometer` history.
- ! Activity flags are **not exclusive**: car at a red light = automotive + stationary; results lag “up to several minutes”.
- ! Shake-to-undo needs no Core Motion: `UIResponder.motionEnded(.motionShake, with:)`. Simulator has no motion hardware.

#### Remember

**“One manager, fused not raw, seconds not Hz, quaternion not Euler — history lives in the coprocessor.”**